AGENT MODE · BATCH 1 OF 5

I built a marketing analytics team that works 24/7, costs $15/month, and answers in 12 seconds.

Here's who's on it — and how I built them in a weekend.

[SCREENSHOT: Terminal Pipeline — All 7 Agents Firing]

Run: python3 main.py → type 'demo' → screenshot the full terminal output

Most marketing leaders I know are understaffed on the analytics side. Not because the budget isn't there. Because the right combination of people — someone who lives in the data, someone who interprets it, someone who knows the creatives inside out, someone who manages budget like a chess game — rarely exists in one team, at one time, at the speed the business actually needs.

> ***So I stopped looking for that team. I built it.***

Over one weekend sprint — 6 to 8 hours of focused building — I assembled a **7-person marketing analytics team**. Except every person is an AI agent with a specific role, a defined expertise, and a clear reporting line. They're always available. They don't have calendar conflicts. They don't need a brief by Wednesday to have slides ready by Friday.

Meet the Team

Orchestrator Agent
The Director

Receives every question, classifies intent, routes to the right specialist, coordinates the team, synthesises the final answer. The only agent the user ever talks to. It does nothing but orchestration — and it does it perfectly.

Data Agent
The Data Engineer

Converts plain-English questions into SQL, runs them against 112,000 rows of marketing performance data in real time, and returns clean structured results. Handles ambiguous time windows. Retries on error. Nobody else touches the database.

Analysis Hub Agent

The Senior Performance Analyst

Takes raw data and produces structured performance insights: ROAS vs target, trend direction (improving / declining / stable), anomaly detection, week-over-week shifts, and objective-aligned evaluation. Uses the more powerful reasoning model because this is where thinking depth matters.

Creative Analyst Agent

The Creative Strategist

Plots CTR curves over the lifetime of each format. Detects the fatigue inflection point — the moment performance starts decaying faster than expected. Identifies format winners by platform. Flags creatives that need refresh before ROAS drops. Most teams find out after the fact. This agent finds it while there is still budget to act.

Optimizer Agent

The Budget Manager

Gives specific budget reallocation recommendations: platform, campaign, SGD amount, reason, projected ROAS impact. Not vague advice — exact numbers, before-and-after comparison, respects minimum spend floors, flags risks.

Dashboard Agent

The Reporting Analyst

Generates a live Streamlit dashboard on demand. 4 tabs: Agent Chat, Performance Dashboard, Creative Health, Data Explorer. WBD-branded design. Always built from live data. Launches in your browser in seconds.

Quality Critic Agent

The Editor

Every insight — before it reaches you — is scored 0–10 across five dimensions: Specificity, Actionability, Accuracy, Completeness, Tone. Score 8+: approved. 6–7: enhanced. Below 6: weakest section rewritten. An AI reviewing AI output. That is where quality control comes from.

[SCREENSHOT: Dashboard — Performance Tab with WBD Logo + KPI Cards]

Run: venv/bin/streamlit run dashboard_app.py → screenshot the Performance tab

The Architecture

The pattern is called **Supervisor / Orchestrator**. One central agent manages the workflow. Specialists do one job each. No agent talks directly to another — everything routes through the Orchestrator. It's explicit, debuggable, and production-friendly.

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■ Routes · Coordinates ■
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How the Team Works Together — A Real Example

Here is what happens when you ask: *"Why is ROAS declining on video creatives?"*

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You → Orchestrator (classifies: creative + performance question)
→ Data Agent (pulls 30-day video creative data — 2.1 seconds)
→ Analysis Hub ("ROAS down 0.6x WoW, CTR declining on 3 assets")
→ Creative Analyst ("3 assets past fatigue threshold — 21+ days on-air")
```

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→ Optimizer ("Reallocate to Carousel formats — projected +0.4x ROAS")
→ Quality Critic ("■ Quality Score: 9/10 — Approved")
→ You (full structured answer)

Total time: 11 seconds.
```

The answer your team would have had by Thursday. In 11 seconds.

[SCREENSHOT: Agent Chat Tab — Pipeline Steps Visible on Right]
Run a query in dashboard Agent Chat tab → screenshot showing pipeline steps panel

Why This Matters Beyond the Technology

This is not really about AI agents. It is about what happens when you **model a team as a system**.

→ Every role has a clear, singular responsibility.
→ Every handoff is explicit — no Slack threads, no 'did you see my message?'
→ Every output has a quality gate before it reaches the decision-maker.
→ The architecture is reusable. Swap the domain knowledge. Keep the pattern.

Building this in AI forced me to think clearly about what each function *actually does*, what inputs it needs, and what good output looks like. That clarity made the AI team better. And it made me think differently about how *any* team should be structured.

Retail. SaaS. E-commerce. Media. FMCG. Swap the agents' domain knowledge. Keep the orchestration pattern. The framework holds.

[SCREENSHOT: Creative Health Tab — Fatigue Heatmap + CTR vs Benchmark]
Navigate to Creative Health tab in dashboard → screenshot both charts

What It's Built With

Layer	Technology	Why
Language	Python 3.12	Stability, ecosystem, tooling
Database	DuckDB 1.5	112K rows, millisecond queries, zero infra
AI (Reasoning)	Claude Sonnet 4.6	Deep analysis, insights, complex reasoning
AI (Fast)	Claude Haiku 4.5	SQL generation, classification, critic scoring
Dashboard	Streamlit + Plotly	Open source, no cloud required, instant deploy

Terminal UI	Rich	Live agent pipeline display with colour + icons
Data	Simulated (112K rows)	4 platforms, 5 business lines, 90 days, SGD

What V1 Doesn't Do Yet (And That's OK)

This is a working V1, not a finished product. The honest limitations:

- Memory doesn't persist across sessions — ask the same thing twice, it starts fresh.
- Complex multi-part questions sometimes need better routing logic.
- The dashboard is regenerated each time, not truly real-time streaming.
- No authentication, user management, or multi-tenant support yet.

But improving a working team is very different from building one from scratch. The foundation is solid. Everything else is iteration.

The Roadmap

Status	Batch	Focus
■ DONE	Batch 1 · Now	7 agents built · Terminal UI · Dashboard V1
■ NEXT	Batch 2 · +2 weeks	Dashboard upgrade · Tabs · Live queries · Creative Health
■ SOON	Batch 3 · +5 weeks	Use Case #2 — same architecture, new industry domain
■ LATER	Batch 4 · +8 weeks	3–4 polished use cases running in parallel
■ FUTURE	Batch 5 · +12 weeks	Company formation · AI micro-agency · Production

I am documenting the full build — architecture, decisions, failures, breakthroughs — in batches as the project grows. **Not when it is finished. Because the journey is the content.**

If you run a marketing team and have a question you have been waiting three days to get answered — drop it in the comments. I will run it through the system and share what comes back.

Follow for Batch 2 — dashboard upgrade, second use case, and what breaks next.

#AIMarketing #MarketingAnalytics #MultiAgentAI #BuildingInPublic #MarketingTech #AIAgents #AgentMode #WeekendProject #LLM #Anthropic

Screenshot Guide — Before You Publish

Take these 4 screenshots and replace the placeholder boxes above before publishing the article.

Screenshot 1 — Terminal Pipeline

```
cd ~/Desktop/marketing-analytics-agent
source venv/bin/activate
python3 main.py
→ Type: demo
→ Let all 5 queries run. Scroll up. Screenshot the full terminal showing agent steps.
```

Screenshot 2 — Dashboard Performance Tab

```
venv/bin/streamlit run dashboard_app.py
→ Navigate to: ■ Performance Dashboard tab
→ Screenshot: WBD logo header + all 6 KPI cards + daily trend chart
```

Screenshot 3 — Agent Chat with Pipeline

```
→ In dashboard, click: ■ Agent Chat tab
→ Ask: 'Which platforms are underperforming vs ROAS target?'
→ Screenshot: chat response + pipeline steps showing on the right
```

Screenshot 4 — Creative Health Tab

```
→ In dashboard, click: ■ Creative Health tab
→ Screenshot: ROAS by format + CTR vs benchmark + fatigue table
```

Tip: Use CMD+SHIFT+4 on Mac to take a precise screenshot. Crop tightly. Dark background screenshots look great on LinkedIn.